

Parallel Corpus Repair for Bhojpuri–Hindi using Bilingual Embeddings and Django Integration

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1 Abstract

Parallel corpora are essential for multilingual NLP tasks but are often noisy and mis-aligned for low-resource languages. This project focuses on repairing Bhojpuri–Hindi (bho–hin) parallel corpora using bilingual embeddings. The system performs corpus cleaning, alignment, embedding training, and dictionary generation. It is further integrated into a Django-based web application. Evaluation is conducted using Precision@k and Mean Reciprocal Rank (MRR).

2 Introduction

Low-resource languages such as Bhojpuri (ISO 639-3: bho) lack high-quality parallel datasets. These datasets often suffer from:

- Sentence misalignment
- Noise and duplication
- Incorrect language mixing

This project uses a bilingual embedding-based approach to repair such corpora and improve cross-lingual alignment.

3 Dataset

The dataset used in this project consists of:

- Source Language: Bhojpuri (bho)
- Target Language: Hindi (hin)
- Format: Parallel text corpus

Preprocessing steps include:

- Indic normalization
- Language identification and filtering
- Tokenization and cleaning

4 Objectives

- Repair noisy parallel corpora
- Train bilingual word embeddings
- Perform cross-lingual alignment
- Generate bilingual dictionaries
- Deploy system using Django APIs

5 Methodology

5.1 Corpus Cleaning and Repair

The corpus is cleaned using language identification and normalization. Repair techniques include:

- Removing incorrect language sentences
- Fixing misaligned sentence pairs
- Correcting segmentation errors

5.2 Embedding Training

FastText is used to train embeddings with:

- 300 dimensions
- Subword modeling for handling morphology

5.3 Cross-Lingual Mapping

Orthogonal Procrustes mapping aligns source and target embeddings into a shared vector space.

5.4 Dictionary Generation

CSLS (Cross-domain Similarity Local Scaling) is used to generate bilingual word mappings.

6 Evaluation

6.1 Metrics

The system is evaluated using alignment and dictionary quality metrics:

- Precision@1
- Precision@5
- Mean Reciprocal Rank (MRR)

6.2 Results

Metric	Value
Precision@1	0.20
Precision@5	0.35
MRR	0.25

6.3 Analysis

- Precision@1 indicates moderate top prediction accuracy.
- Precision@5 shows improved performance when considering multiple candidates.
- MRR reflects that correct translations are often ranked highly.

7 Django Integration

The system is integrated into a Django-based NLP Lab website using REST APIs:

- GET /api/embeddings/translate/
- GET /api/embeddings/similar/
- POST /api/embeddings/batch-translate/

This enables real-time interaction with the embedding system.

8 Applications

- Bilingual dictionary generation
- Cross-lingual NLP tasks
- Low-resource language processing

9 Advantages

- Suitable for low-resource Indic languages
- Handles noisy and misaligned corpora
- Modular and extensible architecture

10 Limitations

- Performance depends on corpus quality
- Limited accuracy compared to large transformer models

11 Future Work

- Incorporate transformer-based embeddings
- Improve alignment using neural approaches
- Extend to additional Indic languages

12 Conclusion

This project demonstrates an effective approach for repairing Bhojpuri–Hindi parallel corpora using bilingual embeddings. The system achieves meaningful alignment results and provides a foundation for further cross-lingual NLP applications.

13 References

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